Flood described in the Bible, had re-formed the surface of the planet many times, reshaping existing mountains, lakes, and rivers and wiping out many plant and animal species. In contrast, followers of uniformitarianism contended that Earth's features were the result of continuous and uniform natural processes of erosion, sedimentation (the depositing of particles carried by fluid flows), and volcanism.

Detailed geological studies demonstrated that neither camp was right. They established that Earth's history has been a process of slow change, punctuated by catastrophic events. The study of glaciers, and the landforms they create, informed these ideas. After observing parallel striations in rocks of the Swiss Alps, German–Swiss geologist Jean de Charpentier (or Johann von Charpentier) postulated that glaciers in the Alps had once been more extensive and had caused the scratches as they moved and their sediment cut into the rock. Geologist Jens Esmark drew similar conclusions in Norway.